



Proof by Induction · Worksheet 1

Core Pure 1 Chapter 8: Proof by Induction. Answer all questions. Show your working clearly. Total: 67 marks. Time: 85 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 8. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- In every proof by induction, $n \in \mathbb{Z}^+$. Set out the four steps clearly: the basis, the assumption, the inductive step and a conclusion sentence.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Induction for summation formulae	20	
B Induction for divisibility	20	
C Induction for matrix powers	22	
Roar-level challenge	5	
Total	67	

Section A Induction for summation formulae

20 marks

1. A proof by induction is to be used to show that $\sum_{r=1}^n r = \frac{1}{2}n(n+1)$. [2]
Show that the result is true for $n = 1$, and write down the assumption made in the inductive step.

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2. Given that $\sum_{r=1}^k (2r-1) = k^2$, show that $\sum_{r=1}^{k+1} (2r-1) = (k+1)^2$. [3]

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3. A student claims that $\sum_{r=1}^n r = \frac{1}{2}(n^2 + n + 2)$. She shows that *if* the formula is true for $n = k$, then it is true for $n = k + 1$. [3]

- (a) Show that her inductive step is correct.
(b) Explain why her claim is nevertheless false.

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4. Prove by induction that [4]

$$\sum_{r=1}^n r(r+1) = \frac{1}{3}n(n+1)(n+2).$$

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5. (a) Prove by induction that, for $n \in \mathbb{Z}^+$,

[5]

$$\sum_{r=1}^n r^3 = \frac{1}{4}n^2(n+1)^2.$$

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(b) Hence show that $\sum_{r=n+1}^{2n} r^3 = \frac{1}{4}n^2(3n+1)(5n+3).$

[3]

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(Total for Question 5 is 8 marks)



Section B Induction for divisibility

20 marks

6. $f(n) = 5^n - 1$. Show that $f(k+1) = 5f(k) + 4$.

[2]

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7. Using the result of Question 6, prove by induction that $5^n - 1$ is divisible by 4 for all $n \in \mathbb{Z}^+$.

[3]

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8. Prove by induction that $3^{2n} + 7$ is divisible by 8 for all $n \in \mathbb{Z}^+$.

[4]

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9. Prove by induction that $n^3 + 2n$ is divisible by 3 for all $n \in \mathbb{Z}^+$.

[4]

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10. $f(n) = 3^{4n} + 2^{4n+2}$.

(a) Show that $f(k+1) - 16f(k) = 65 \times 3^{4k}$. [2]

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(b) Hence prove by induction that $f(n)$ is divisible by 5 for all $n \in \mathbb{Z}^+$. [4]

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(c) Given that $f(1) = 145 = 5 \times 29$, determine whether $f(n)$ is divisible by 29 for all $n \in \mathbb{Z}^+$. [1]

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(Total for Question 10 is 7 marks)



Section C Induction for matrix powers

22 marks

11. $\mathbf{M} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$. Find $\mathbf{M}^2$ and $\mathbf{M}^3$, and hence suggest a formula for $\mathbf{M}^n$. [2]

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12. For the matrix $\mathbf{M}$ in Question 11, prove by induction that $\mathbf{M}^n = \begin{pmatrix} 1 & 2n \\ 0 & 1 \end{pmatrix}$ for all $n \in \mathbb{Z}^+$. [3]

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13. Prove by induction that, for all $n \in \mathbb{Z}^+$, [4]

$$\begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}^n = \begin{pmatrix} 2^n & 0 \\ 2^n - 1 & 1 \end{pmatrix}.$$

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14. Prove by induction that, for all $n \in \mathbb{Z}^+$,

[4]

$$\begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}^n = \begin{pmatrix} 1+2n & -4n \\ n & 1-2n \end{pmatrix}.$$

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15. $\mathbf{M} = \begin{pmatrix} 5 & -8 \\ 2 & -3 \end{pmatrix}$.

(a) Prove by induction that, for all $n \in \mathbb{Z}^+$,

[5]

$$\mathbf{M}^n = \begin{pmatrix} 1+4n & -8n \\ 2n & 1-4n \end{pmatrix}.$$

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(b) Show that $\det(\mathbf{M}^n) = 1$ for all $n \in \mathbb{Z}^+$, and interpret this geometrically.

[2]

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(c) Find $\mathbf{M}^{-1}$, and show that the formula in part (a) also holds when $n = -1$.

[2]

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(Total for Question 15 is 9 marks)



ROAR-LEVEL CHALLENGE

5 marks

16. Prove by induction that, for all $n \in \mathbb{Z}^+$,

[5]

$$\begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix}^n = \begin{pmatrix} 2^{n+1} - 1 & 2 - 2^{n+1} \\ 2^n - 1 & 2 - 2^n \end{pmatrix}.$$

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END OF WORKSHEET

Total: 67 marks